

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)
1	9	33.15
2	13	33.33
3	5	33.50
4	7	33.55
5	12	33.73
6	6	33.86
7	4	33.90
8	5	34.15
9	3	34.90

- a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
- b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

H_0 : There's no correlation.

H_a : There's a negative correlation.

$\alpha = 0.05$.

$r_{9, 0.05} = -0.6$.

RR: $\{r_s \mid r_s < -0.6\}$

$-0.7375 < -0.6 \rightarrow \text{reject } H_0$

There is sufficient evidence to say that there is negative correlation between years of competitive running experience and 10-kilometer finish time.

Rank X	Rank Y	$d = x_i - y_i$
7	1	6
9	2	7
3.5	3	0.5
6	4	2
8	5	3
5	6	-1
2	7	-5
3.5	8	-4.5
1	9	-8

$$r_{s, \text{STAT}} = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$$r_s^* = 1 - \frac{6(6^2 + 7^2 + 0.5^2 + 2^2 + 3^2 + (-1)^2 + (-5)^2 + (-4.5)^2 + (-8)^2)}{9(81 - 1)}$$

$$= -0.7375$$